Veristake

Insurance claims, verified by economics.

Whitepaper v2.0 — Product Revision

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Live demo: https://veristake-demo.vercel.app

Investor TLDR

Veristake is the verification layer for disputed insurance claims. Health and auto carriers keep underwriting authority and reserves, while Veristake routes contested claim packets to credentialed verifiers who stake capital, build reputation, and can be penalized for reckless decisions.

The result is faster, auditable, bias-resistant claim resolution without making Veristake an insurer.

Executive Summary

Veristake is a disputed-claim verification protocol layer for licensed health and auto carriers. It is designed for cases where trust breaks down: denied appeals, delayed payouts, complex evidence packets, suspected fraud, and other high-friction claims that need outside adjudication capacity, bias-resistant review, and an auditable trail without handing underwriting authority to a crypto-native insurance product.

The final Phase 2 product now demonstrates this end-to-end: a public landing page for non-crypto insurance executives, a live Base Sepolia dashboard reading production-grade testnet contracts, and a no-wallet demo surface where a visitor can experience claimant, carrier, and verifier flows in minutes.

The core mechanism is simple. Credentialed verifiers put VST at risk, review structured claim packets, commit and reveal votes, and build a reputation history. Accurate review is rewarded; provably wrong or reckless review can be slashed. But Veristake is not raw majority rule: verifier influence should reflect credential scope, historical accuracy, reputation, stake, and case fit, with appeal and arbiter correction for hard cases.

- Health and auto are the only launch domains in the current product.
- Veristake is not intended to process every routine claim; the first wedge is disputed, high-friction, or fraud-sensitive claims where external review has high value density.
- The dashboard reads the production-grade Base Sepolia deployment; visitors do not write to it.
- Demo writes are intended for a Tenderly Base Sepolia virtual testnet with compressed timing and backend-controlled verifier wallets.
- No personally identifiable claim evidence belongs on-chain; on-chain data is metadata, state, stake, votes, reputation, reserve, and payout movement.

1. The Problem

1.1 Denials Are Not a Marginal Edge Case

Insurance executives already know the operational tension: claims teams must control fraud and leakage, yet policyholders and regulators punish opaque denials. Veristake focuses on this tension instead of trying to become a carrier.

Statistic	Why it matters	Source used in product
80.7%	Medicare Advantage prior-authorization denials overturned on appeal in 2024.	KFF, Medicare Advantage prior authorization determinations in 2024.
19%	In-network ACA marketplace health claims denied in 2024.	KFF, Claims Denials and Appeals in ACA Marketplace Plans in 2024.
37%	Out-of-network ACA marketplace health claims denied in 2024.	KFF, same source.
34%	ACA marketplace denial appeals reversed by insurers in 2024.	KFF, same source.
Less than 1%	Denied ACA marketplace claims that patients formally appealed in 2024.	KFF, same source.
65.2%	Closed insurance complaints involving claim handling in 2024.	NAIC Consumer Information Source data, summarized by ValuePenguin.
+31.6%	Auto insurance complaint growth from 2021 to 2024.	NAIC Consumer Information Source data, summarized by ValuePenguin.
29,734	Auto insurance complaints in 2024.	NAIC Consumer Information Source data, summarized by ValuePenguin.

1.2 The Incentive Conflict

Traditional claim handling is centralized inside the carrier or its vendors. That makes sense legally, but it creates a credibility problem: the same institution that funds the reserve often controls the first denial, the first appeal path, and the evidence record. Automation alone does not solve this. A faster denial is not a fairer denial.

The product thesis is that the missing layer is not another AI adjuster. It is economic accountability around human and specialist review: outside verifiers must have something to lose when they ignore evidence, and something to gain when they correctly challenge consensus.

2. What Veristake Is

Item	Current product position
Category	Verification protocol layer for licensed insurance carriers.
Launch domains	Health and auto only. Property, cyber, and other markets are intentionally out of scope for this phase.
Buyer	US health carriers, US auto carriers, brokers, reinsurers, and insurtech product leads.
Visitor promise	Arrive, understand, try a no-wallet demo, watch a claim resolve, and leave with an outcome that can be shown internally.
Legal posture	Software infrastructure. Not an insurance carrier; does not underwrite, sell, or adjudicate policies independently.

2.1 Carrier-Friendly by Design

- The carrier keeps underwriting authority, pricing, policy language, compliance obligations, and customer relationship.
- Payouts come from carrier-funded reserves, not from Veristake balance-sheet risk.
- Verifier pools can be limited by domain, credentials, geography, or policy program.
- The audit trail can support internal quality review, regulator response, reinsurer reporting, and broker transparency.

2.2 Why Disputed Claims First

Veristake should not sit in front of every claim. Routine, low-dollar, low-dispute claims should continue moving through carrier systems, automation, and ordinary adjuster workflows. External verifier review has a cost, so it belongs where the incremental trust, auditability, and fraud-control value is highest.

• Cost efficiency: outside review is reserved for appeals, complex evidence packets, suspected fraud, delayed payouts, or large disputes rather than commodity processing.

• Regulatory positioning: Veristake remains a verification layer inside a licensed carrier pipeline, not a TPA, claims core, or insurer substitute.

• Value density: disputed claims are where claimants distrust carrier judgment and carriers need independent evidence review without surrendering reserves or policy authority.

3. Product Status

Item	Current product position
Live site	https://veristake-demo.vercel.app
Public dashboard	Reads production-grade Base Sepolia contract data from deployment block 41,708,833.
Seeded dashboard state	5 claims, 2 carrier registrations, 1 slashing event, and 1,950 VST staked as demonstration data.
Hero proof asset	Narrated 90-second walkthrough video with poster art and captions.

Item	Current product position
Operational endpoints	Health endpoint, sitemap, robots, OG metadata, legal placeholder, analytics hooks, and error-tracking hooks are implemented.

Important limitation

The live dashboard is real Base Sepolia testnet data. The full embedded-wallet demo requires founder-provided Tenderly, Privy, faucet, and session-store secrets before it becomes fully transactional in production.

4. Protocol Architecture

4.1 End-to-End Claim Flow

- Carrier registers itself and a policy through CarrierGateway.
- Carrier funds a payout reserve for that policy.
- Carrier or claimant routes a disputed, high-friction, or fraud-sensitive claim packet to ClaimRegistry.
- Eligible verifiers stake VST and enter the domain pool.
- Voting runs through commit, reveal, resolution, optional appeal, and finalization.
- CarrierGateway releases approved or partial payouts from the carrier reserve.
- Reputation and slashing contracts update verifier accuracy and penalties.

4.2 Contract Modules

Module	Role
ClaimRegistry	Registers claim metadata, domain, requested amount, policy link, and lifecycle state.
Voting	Runs commit/reveal voting, challenge windows, outcomes, and production/demo parameter windows.
VerifierStaking	Manages VST stake, domain pool registration, and verifier participation.
SoulboundReputation	Tracks historical verifier accuracy and supports accuracy-weighted trust.
CarrierGateway	Registers carriers and policies, funds carrier payout reserves, and releases approved payouts.
Slashing	Applies disagreement tolerance, supermajority thresholds, and bond redistribution.
ArbiterEscalation	Supports appealed or ambiguous decisions through a second adjudication layer.
VST	Verifier Staking Token used for demo/protocol staking and slashing-accountability mechanics.
TimelockController	Owns production administration after setup so protocol changes are delayed and auditable.

4.3 Dual Deployment Model

Veristake separates proof and speed. The public dashboard reads a production-grade Base Sepolia deployment with default long windows. The guided demo writes to a Tenderly Base Sepolia virtual testnet where timing can be compressed for sales demos.

Item	Current product position
Production-grade testnet	Real Base Sepolia. Used for public read-only dashboard metrics and audit-clean testnet history.
Demo sandbox	Tenderly virtual Base Sepolia. Used for visitor actions, backend-controlled demo verifiers, faucet balances, and time-warped claim flows.
Why two deployments	Keeps the real testnet deployment credible while allowing a prospect to finish a claim flow in a few minutes.

5. Incentive Design

5.1 Staking and Voting

Verifiers stake VST to participate in a domain. Claim review uses commit/reveal voting so participants cannot simply copy early votes. Once votes resolve, the system updates claim state, reputation, and payout eligibility.

5.2 Reputation

Soulbound reputation tracks accuracy over time. In the current deployment, reputation is a protocol primitive rather than a consumer-facing identity product: it lets the network distinguish durable expertise from one-off participation.

5.3 Slashing and Dissent

Slashing is reserved for serious divergence from the final adjudicated result, not honest disagreement in a gray area. The launch demo intentionally shows this distinction: a health verifier can dissent on an ambiguous ER appeal without being slashed, but approving clear duplicate-billing fraud can trigger a 50 VST penalty.

Contrarian rewards are central to the design. Veristake should not punish a reviewer for being in the minority if the minority is ultimately right. The goal is not blind consensus. The goal is accountable accuracy.

5.4 Accuracy Is Not Raw Majority Rule

A verifier market only works if it resists herd mentality. Veristake therefore should not be understood as community arbitration where the largest group automatically determines truth. The protocol needs a professional weighting model: credential scope, domain relevance, historical accuracy, reputation, stake at risk, and conflict signals should all affect influence.

Economic incentives are necessary but not sufficient. Stake and slashing make reckless review costly, but they do not by themselves establish medical necessity, collision causation, fraud, or policy interpretation. Accuracy also depends on evidence standards, verifier credentialing, calibrated reputation, appeal rights, and, for hard cases, arbiter-panel correction.

This is why honest dissent should be protected. A minority verifier who is later vindicated should be eligible for contrarian rewards, while a verifier who follows the majority into a clearly wrong outcome should still face reputation loss or slashing if the error is reckless and out of tolerance.

6. Demo Experience

The demo is built for insurance executives who do not want a wallet, testnet ETH, seed phrases, or token onboarding. The target path is: understand the value proposition, pick a persona, see a claim packet, watch verifier activity, and finish with a visible resolution.

Persona	Scenario	Expected proof moment
Health claimant	**ER visit denial appeal**	$4,820 chest-pain ER visit denied as not medically necessary; verifier pool approves 4-1 and payout releases from carrier reserve.
Health verifier	**Three claims, one fraud**	Visitor reviews an annual physical, the ER appeal, and duplicate-billed PT sessions; wrong fraud approval triggers a 50 VST slash.
Auto claimant	**Rear-end collision delayed payout**	$7,140 repair estimate vs. $3,200 insurer offer; verifier median resolves at a partial $6,800 payout.
Auto carrier	**Pacific Mutual onboarding**	Carrier registers, creates an auto-collision policy, funds a $10k reserve, and watches three claims resolve from reserve.
Auto verifier	**Staged-collision fraud pattern**	Visitor reviews a clean fender-bender, the rear-end claim, and a fraud pattern with repeated VIN/body-shop/attorney signals.

6.1 Persona Flows

- Carrier: register Pacific Mutual, register an auto policy, fund a reserve, and watch three claims resolve.
- Claimant: submit a health ER denial appeal or auto collision dispute and watch the resolution panel update.
- Verifier: stake demo VST, review three claims, vote, and see reward, accuracy, and slashing outcomes.

7. Competitive Landscape

Veristake is not positioned as another crypto insurance pool. Its differentiation is the combination of human/specialist review, slashing-backed accountability, carrier-reserve payouts, and a regulated-integration posture.

Feature	Veristake	Etherisc	Nexus Mutual	Kleros	Status quo
Human review	Yes; credentialed domain verifiers	Limited; mostly parametric	DAO/member assessment	Yes; generalized arbitration	Yes; internal adjusters
Economic penalties	Yes; VST slashing	No verifier-slashing layer	Capital-at-risk mutual model	Yes; juror incentives	Employment/legal controls
Contrarian rewards	Yes; accurate dissent can be rewarded	No	Limited	No direct carrier claim design	No formal market reward
Carrier-friendly	Designed as protocol layer for licensed carriers	Mostly decentralized insurance products	Mutual membership model	General dispute protocol	Native workflow but opaque
Regulated-friendly	Carrier keeps underwriting authority and reserves	Limited	Limited	Limited	High, but biased by incentives
Privacy-aware	No PII on-chain; private evidence surfaces planned	Basic	Basic	Case-dependent	Internal systems only

8. Privacy, Compliance, and Legal Posture

- Veristake is software infrastructure, not an insurance carrier.
- The carrier remains the regulated party responsible for policies, underwriting, and claims obligations.
- Claim evidence should be stored off-chain with access control; on-chain state should avoid PII.
- Credentialing, KYC, TEEs, ZK proofs, and formal legal review are roadmap items, not claims of completed production compliance.
- The public legal page is explicitly a draft pending founder and counsel review.

9. Business Model

The commercial model should stay simple for early carrier conversations. Veristake sells access to a verification layer: software integration, domain-specific verifier liquidity, and auditable claim-resolution infrastructure.

- Enterprise protocol licensing or SaaS fees for carrier integration.
- Claim review fees paid by carriers, brokers, or program administrators.
- Optional treasury share from protocol fees, subject to legal and tax review.
- VST supports verifier staking and accountability; it should not be marketed as consumer insurance coverage.

10. Roadmap

Status	Milestone	Deliverable
Completed	Phase 1 contracts	Foundry system with registry, voting, staking, reputation, slashing, carrier reserve, and deployment scripts.
Completed	Phase 2 demo surface	Next.js site, sourced landing page, three persona flows, screenshots, narrated walkthrough, legal/footer/metadata polish.
Completed	Production-grade testnet dashboard	Base Sepolia deployment seeded with 5 claims, 2 carriers, 1 slashing event, and 1,950 VST staked for live read-only metrics.
Next	Tenderly/Privy full demo activation	Founder-provided Tenderly and Privy secrets enable real embedded-wallet demo writes in the sandbox.
Next	Pilot integration package	API wrapper, carrier onboarding docs, compliance questionnaire, and sandbox policy templates for health and auto carriers.
Later	Privacy hardening	Credentialing, evidence redaction, optional TEE/ZK proof paths, and formal security/legal review.

11. Team

Oscar Li - Founder. Sole creator and developer of Veristake; leads product direction, protocol design, smart-contract coordination, compliance framing, deployment workflow, and external strategy.

Advisory needs: insurance regulatory counsel, health billing reviewers, auto claims specialists, actuarial/reinsurance advisors, and privacy/security reviewers.

12. Investor and Carrier FAQ

Is Veristake an insurance carrier?

No. Veristake is software infrastructure for verification. Licensed carriers keep policies, reserves, underwriting authority, and legal obligations.

Why not submit every claim?

Because most claims are routine, low-friction, and already well suited to existing carrier systems. Veristake is designed for disputed appeals, large or delayed payouts, complex evidence, suspected fraud, and grey-area cases where independent review creates more value than it costs.

Why would a carrier use it?

To add independent, economically accountable review for disputed or high-friction claims without building a verifier market from scratch.

Why would verifiers act honestly?

They stake VST, build reputation, and face slashing for clearly wrong or reckless review. Correct minority review can be rewarded.

Why health and auto first?

The sourced denial, appeal, and complaint data make the trust problem easy to explain, and the claim packets can be structured enough for a controlled demo.

What is live today?

A deployed site, production-grade Base Sepolia dashboard, seeded protocol metrics, contract system, screenshot/video assets, and setup/deploy automation.

What is still pending?

Full Tenderly/Privy production secrets, formal legal review, custom domain, real carrier logos, and pilot customer validation.

13. References

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- Veristake demo site. https://veristake-demo.vercel.app
- Veristake public dashboard. https://veristake-demo.vercel.app/dashboard
- Veristake repository. https://github.com/oscar174831/veristake

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